Quirk plane search: efficiencies, uncertainties, and Figures 2–5

Comparison with Knapen, Lou, Papucci, and Setford, arXiv:1708.02243

Abstract

This note reports a 2000-event electroweak sample and a 2000-event colored sample of the quirk plane search. Every cut is counted sequentially. Each efficiency is given with its binomial standard error and a Wilson 95% confidence interval, and is compared with Table I of the paper. Figures 2–5 of the paper are reproduced for these benchmarks: the internal angular momentum and separation, the rotation of the quirk plane, one event projected onto its reconstructed plane, and the distributions of the thickness Δs and the width Δw.

1 Samples

Both samples use the straight-string equation of motion, a 2 T solenoid, and the eight-layer barrel. The electroweak sample has mass $m_Q = 800\,\text{GeV}$. The colored sample has mass $m_Q = 1800\,\text{GeV}$. Both use the string scale $\Lambda = 4\,\text{keV}$. Each sample contains $N_{\text{gen}} = 2000$ events, generated on 16 cores. The four-momenta come from the built-in sampler. MadGraph and Pythia were not used.

2 Efficiency definitions

An event is counted at a cut only if it has passed every earlier cut. The three efficiencies in the paper are ratios of those counts,

$$\varepsilon_{\text{trig}} = \frac{N_{\text{trig}}}{N_{\text{gen}}}, \qquad N_{\text{trig}} = \#\{p_T > 200\,\text{GeV}\}, \tag{1}$$

$$\varepsilon_{\text{fid}} = \frac{N_{\text{fid}}}{N_{\text{trig}}}, \tag{2}$$

$$\varepsilon_{\text{reco}} = \frac{N_{\text{reco}}}{N_{\text{fid}}}, \tag{3}$$

$$\varepsilon = \varepsilon_{\text{trig}}\,\varepsilon_{\text{fid}}\,\varepsilon_{\text{reco}} = \frac{N_{\text{reco}}}{N_{\text{gen}}}. \tag{4}$$

The fiducial count N_{fid} requires both quirks to intersect all eight barrel layers, with every hit time below 25 ns. The reconstruction count N_{reco} is the number of events that pass the final hit-pattern cut. Because the flow is sequential, that count already requires the trigger, the fiducial requirement, the seed windows, the factor-of-three growth test, $\Delta w < 1\,\text{cm}$, and $\Delta s < 0.01\,\text{cm}$.

The individual cuts, in order, are:

1. **Generated.** Every simulated event.
2. **Trigger.** Pair transverse momentum above 200 GeV.
3. **Fiducial.** Both quirks hit all eight layers within 25 ns.
4. **Layer-8 seed.** A hit pair on layer 8 with $\Delta\phi < 0.1$ and $\Delta z < 2\,\text{cm}$.

5. **Layer-7 seed.** A hit pair on layer 7 in the same window.

6. **Seed prefilter.** The two pairs agree within 0.5 rad and 80 mm. This removes combinations that cannot satisfy the 1 cm width cut.

7. **Seed thickness.** The four seed hits have $\Delta s < 0.05\,\mathrm{cm}$.

8. **Seed width.** The four seed hits have $\Delta w < 1\,\mathrm{cm}$.

9. **Forward direction.** Every seed hit satisfies $\boldsymbol{n}_3 \cdot \boldsymbol{x} > 0$.

10. **Inward walk.** From layer 6 down to layer 1, a hit is added only when both Δs and Δw stay within a factor of three of their current values, and the finished plane has $\Delta w < 1\,\mathrm{cm}$.

11. **Final thickness.** The finished plane has $\Delta s < 0.01\,\mathrm{cm}$.

12. **Hit pattern.** Seven layers contain two hits and the remaining layer contains at least one.

3 Wilson intervals

Each efficiency is a binomial proportion k/n, with estimate $\hat{p} = k/n$. The standard error is

$$\sigma = \sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}}. \tag{5}$$

The 95% interval is the Wilson score interval. With $z = 1.960$,

$$\begin{aligned} \hat{p}_{\mathrm{W}} &= \left(\hat{p} + \frac{z^2}{2n}\right)\frac{1}{1+z^2/n}, \\ \delta &= \frac{z}{1+z^2/n}\sqrt{\frac{\hat{p}\,(1-\hat{p})}{n} + \frac{z^2}{4n^2}}, \end{aligned} \tag{6}$$

and the interval is $\hat{p}_{\mathrm{W}} \pm \delta$, clipped to $[0, 1]$. For a conditional cut, k is the number of events that pass the cut and n is the number that reached it.

4 Electroweak sample

Mass 800 GeV, string scale 4 keV, $N_{\mathrm{gen}} = 2000$.

Table 1: Sequential cut flow for the electroweak sample. The efficiency is k/n, where n is the number of events that reached the cut.

Cut	N_{pass}	N_{in}	Efficiency	Std. error	Wilson 95% CI
Generated	2000	2000	1.000	0.000	[1.000, 1.000]
Trigger	569	2000	0.284	0.010	[0.265, 0.305]
Fiducial	549	569	0.965	0.008	[0.946, 0.977]
Layer-8 seed	515	549	0.938	0.010	[0.915, 0.955]
Layer-7 seed	503	515	0.977	0.007	[0.960, 0.987]
Seed prefilter	481	503	0.956	0.009	[0.935, 0.971]
Seed Δs	481	481	1.000	0.000	[0.992, 1.000]
Seed Δw	480	481	0.998	0.002	[0.988, 1.000]
Forward	480	480	1.000	0.000	[0.992, 1.000]
Inward walk	480	480	1.000	0.000	[0.992, 1.000]
Final Δs	480	480	1.000	0.000	[0.992, 1.000]
Hit pattern	338	480	0.704	0.021	[0.662, 0.743]

Table 2: Electroweak efficiencies as defined in Eqs. (1)–(4).

Efficiency	Count	Estimate	Std. error	Wilson 95% CI
$\varepsilon_{\text{trig}}$	569/2000	0.284	0.010	[0.265, 0.305]
ε_{fid}	549/569	0.965	0.008	[0.946, 0.977]
$\varepsilon_{\text{reco}}$	338/549	0.616	0.021	[0.574, 0.655]
ε	338/2000	0.169	0.008	[0.153, 0.186]

The last row of Table 1 is not $\varepsilon_{\text{reco}}$. It is the hit-pattern efficiency among events that already passed the final Δs cut, $338/480 = 0.704$. The paper's reconstruction efficiency uses the fiducial sample as the denominator, $338/549 = 0.616$.

5 Colored sample

Mass 1800 GeV, string scale 4 keV, $N_{\text{gen}} = 2000$. The 30% charged quirk-hadron fraction of the paper is applied as a weight on the yield, not as an extra cut in the efficiency.

Table 3: Sequential cut flow for the colored sample.

Cut	N_{pass}	N_{in}	Efficiency	Std. error	Wilson 95% CI
Generated	2000	2000	1.000	0.000	[1.000, 1.000]
Trigger	895	2000	0.448	0.011	[0.426, 0.469]
Fiducial	556	895	0.621	0.016	[0.589, 0.652]
Layer-8 seed	533	556	0.959	0.008	[0.939, 0.972]
Layer-7 seed	519	533	0.974	0.007	[0.956, 0.984]
Seed prefilter	518	519	0.998	0.002	[0.989, 1.000]
Seed Δs	518	518	1.000	0.000	[0.993, 1.000]
Seed Δw	510	518	0.985	0.005	[0.970, 0.992]
Forward	510	510	1.000	0.000	[0.993, 1.000]
Inward walk	510	510	1.000	0.000	[0.993, 1.000]
Final Δs	510	510	1.000	0.000	[0.993, 1.000]
Hit pattern	340	510	0.667	0.021	[0.625, 0.706]

Table 4: Colored efficiencies as defined in Eqs. (1)–(4).

Efficiency	Count	Estimate	Std. error	Wilson 95% CI
$\varepsilon_{\text{trig}}$	895/2000	0.448	0.011	[0.426, 0.469]
ε_{fid}	556/895	0.621	0.016	[0.589, 0.652]
$\varepsilon_{\text{reco}}$	340/556	0.612	0.021	[0.570, 0.651]
ε	340/2000	0.170	0.008	[0.154, 0.187]

6 Comparison with Table I

Table I of arXiv:1708.02243 gives $\varepsilon_{\text{trig}} = 0.10$ and $\varepsilon_{\text{fid}} = 0.28$ for the 800 GeV electroweak benchmark, independent of Λ, and $\varepsilon_{\text{reco}} = 0.72$ at $\Lambda = 4$ keV. For the 1800 GeV colored benchmark it gives $\varepsilon_{\text{trig}} = 0.24$ and $\varepsilon_{\text{fid}} = 0.28$. It does not list a colored reconstruction efficiency at 4 keV. The neighboring colored entries are $\varepsilon_{\text{reco}} = 0.59$ at 3 keV and $\varepsilon_{\text{reco}} = 0.74$ at 5 keV. The paper does not quote uncertainties on these entries.

Table 5: Table I compared with this work. A Table I value is marked outside the interval when it does not lie in the Wilson 95% interval of the corresponding measurement.

Sample	Efficiency	Table I	This work	Wilson 95% CI	Table I
Electroweak	$\varepsilon_{\mathrm{trig}}$	0.10	0.284	[0.265, 0.305]	outside
Electroweak	$\varepsilon_{\mathrm{fid}}$	0.28	0.965	[0.946, 0.977]	outside
Electroweak	$\varepsilon_{\mathrm{reco}}$	0.72	0.616	[0.574, 0.655]	outside
Colored	$\varepsilon_{\mathrm{trig}}$	0.24	0.448	[0.426, 0.469]	outside
Colored	$\varepsilon_{\mathrm{fid}}$	0.28	0.621	[0.589, 0.652]	outside
Colored	$\varepsilon_{\mathrm{reco}}$	—	0.612	[0.570, 0.651]	not tabulated

The trigger and fiducial efficiencies are properties of the momentum distribution. Table I uses MadGraph and Pythia. This run uses the built-in sampler, and those two efficiencies differ from the paper by much more than the statistical uncertainty. The reconstruction efficiency uses the same plane cuts. For the electroweak sample, Table I gives 0.72 and this measurement is 0.616 ± 0.021, so 0.72 lies outside the Wilson interval. The colored reconstruction efficiency at 4 keV, 0.612 ± 0.021, lies between the paper's 3 keV and 5 keV entries.

7 Figures

The four figures below correspond to Figures 2–5 of the paper. Figures 2–4 are single representative events at the benchmark masses, in the same 2 T field. Figure 5 uses the 2000-event samples.

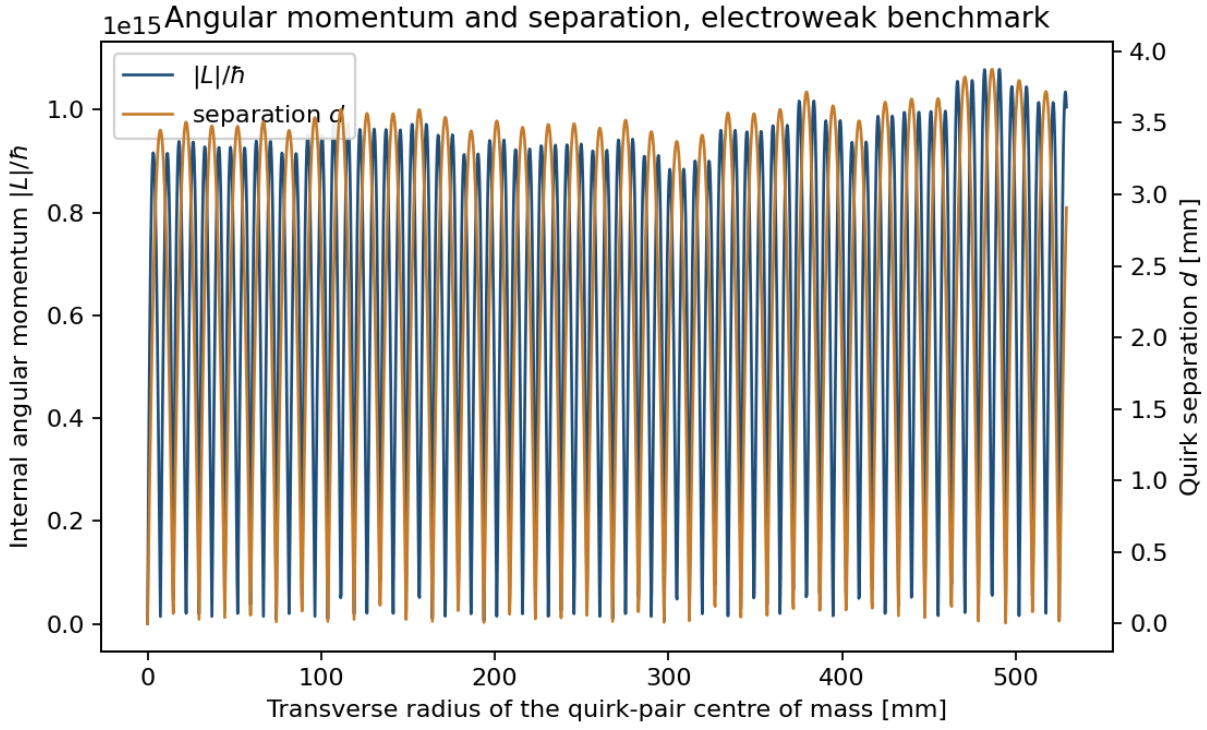


Figure 2: Internal angular momentum $|L|/\hbar$ and quirk separation d versus the transverse radius of the pair centre of mass, for one electroweak event at 800 GeV and 4 keV. On this trajectory $|L|/\hbar$ reaches 1.08×10^{15}, and d oscillates between 0.001 mm and 3.87 mm. This is the same pair of quantities as Figure 2 of the paper.

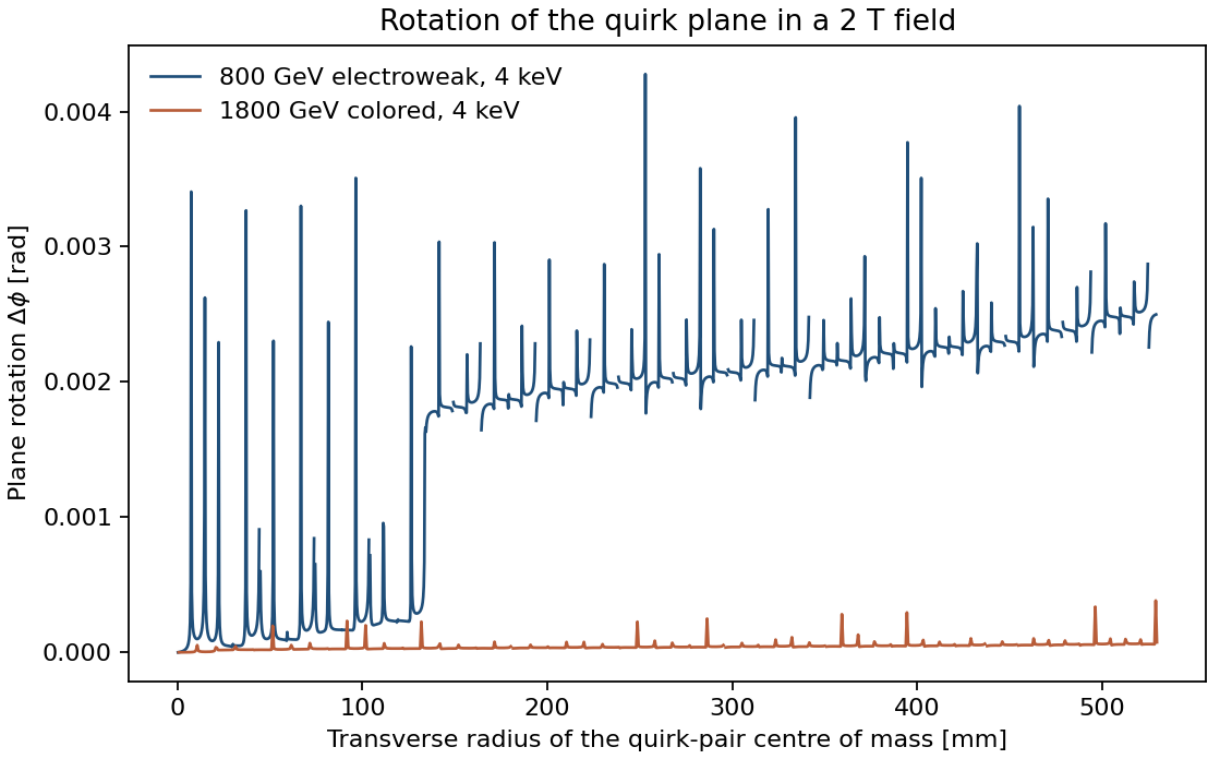


Figure 3: Angle $\Delta\phi$ between the instantaneous quirk plane and the plane at the start of the trajectory. The electroweak event reaches 4.3×10^{-3} rad. The colored event, at 1800 GeV and 4 keV, reaches 3.8×10^{-4} rad. As in Figure 3 of the paper, the rotation remains small compared with the tracker resolution.

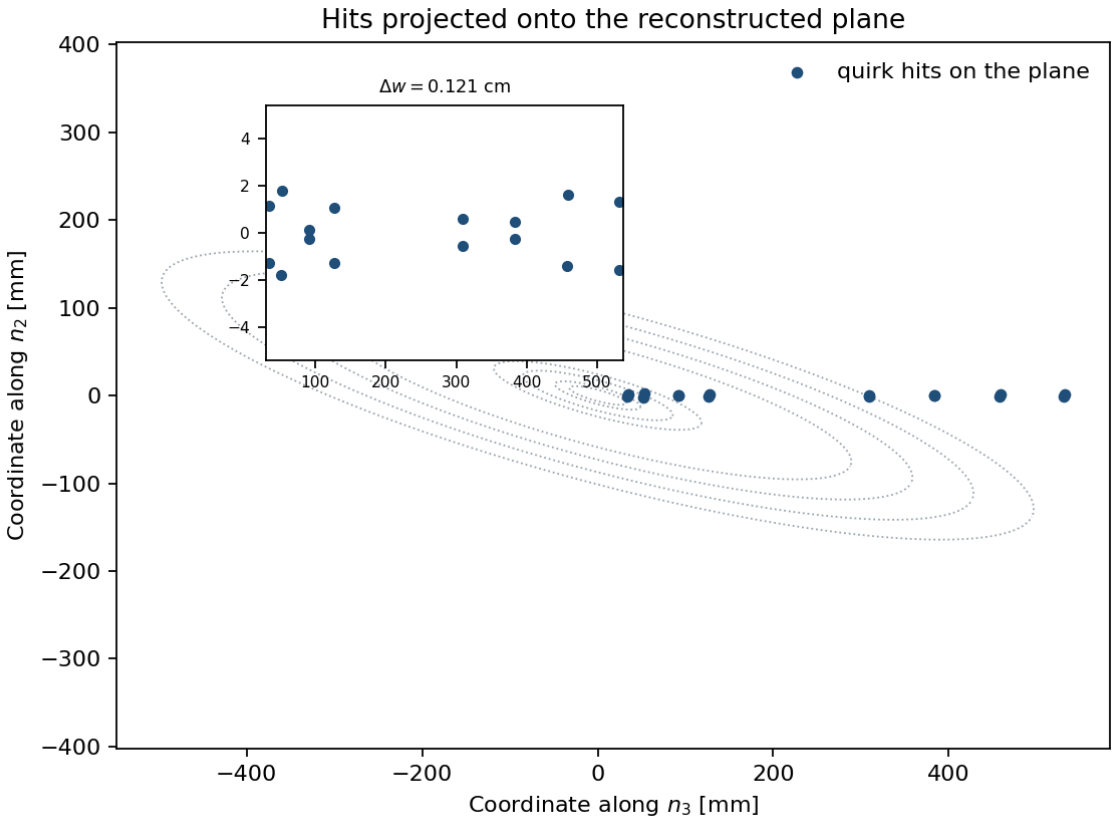


Figure 4: Hits of one electroweak event projected onto the reconstructed $(\boldsymbol{n}_3, \boldsymbol{n}_2)$ plane. The dotted curves are the eight barrel layers. The fit gives $\Delta s = 0.0027\,\text{cm}$ and $\Delta w = 0.121\,\text{cm}$, from 16 hits. The inset shows the strip, as in Figure 4 of the paper.

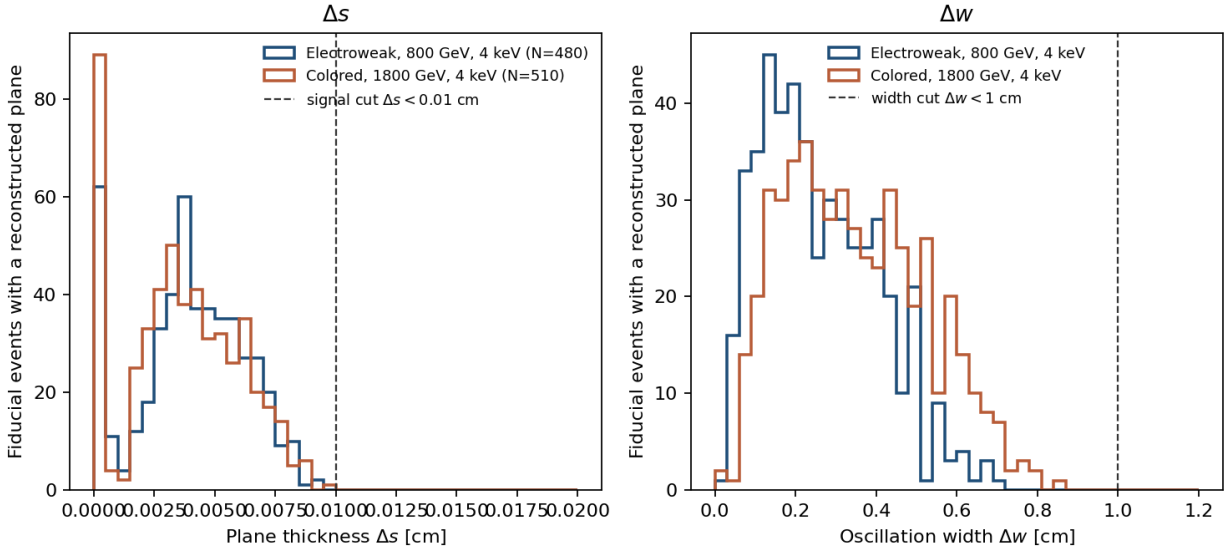


Figure 5: Thickness Δs and width Δw for fiducial events in which the plane finder returned a plane, before the hit-pattern requirement. The dashed line on the left is the signal cut $\Delta s < 0.01\,\text{cm}$. The dashed line on the right is $\Delta w < 1\,\text{cm}$. There are 480 electroweak events and 510 colored events in the histograms. Figure 5 of the paper also shows the pileup and photon-conversion background, which was not generated in this run, and its colored curve is at 2 keV rather than 4 keV.